

LAB 3 - Configure a semantic model

Duration - 45 minutes

Lab story

In this lab, you'll commence developing the data model. It will involve creating **relationships** between tables, and then configuring table and column properties to improve the friendliness and usability of the data model. You'll also create hierarchies and quick measures.

In this lab, you learn how to:

- Create model relationships.
- Configure table and column properties.
- Create hierarchies.
- Create quick measures.
- Configure a many-to-many relationship.

Get started

To complete this exercise, first open the 03-model-data folder.

Open the 03-Starter-Sales Analysis.pbix file

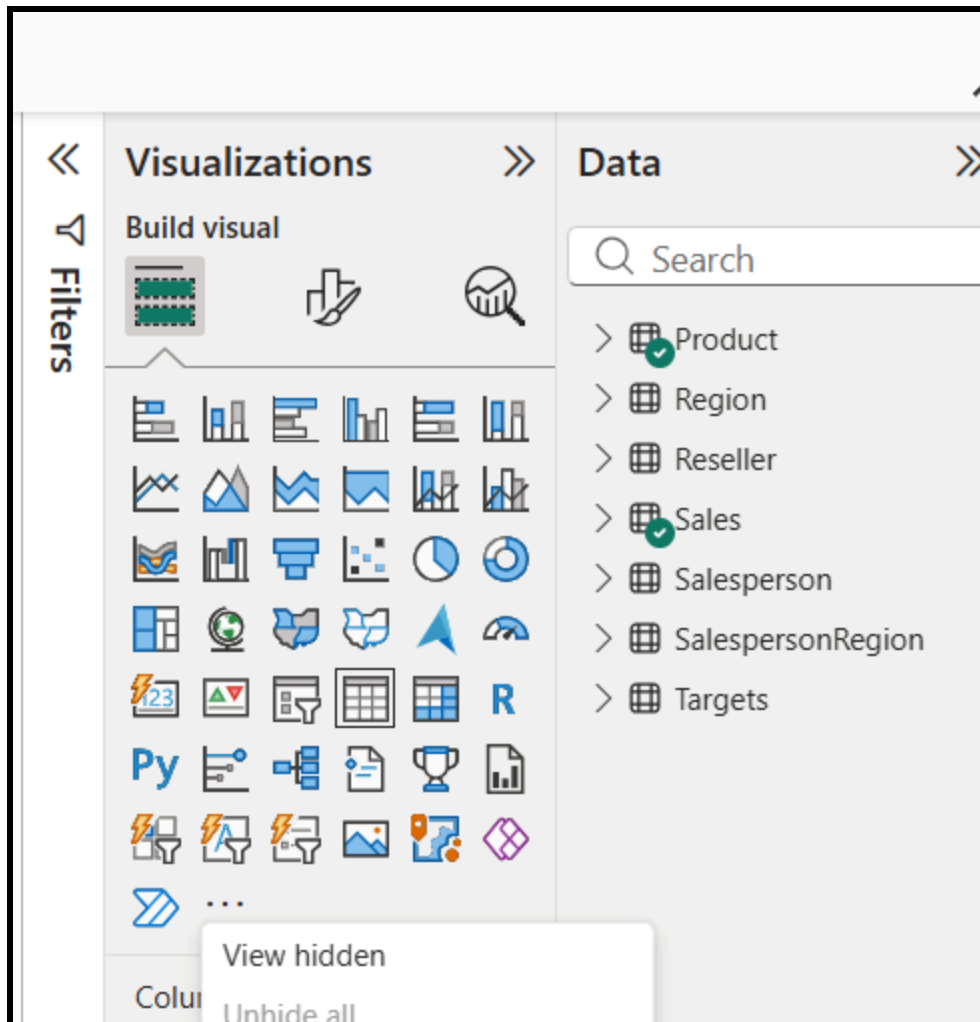
Note: You may see a sign-in dialog as the file loads. Select Cancel to dismiss the sign-in dialog. Close any other informational windows. Select Apply Later, if prompted to apply changes.

Create model relationships

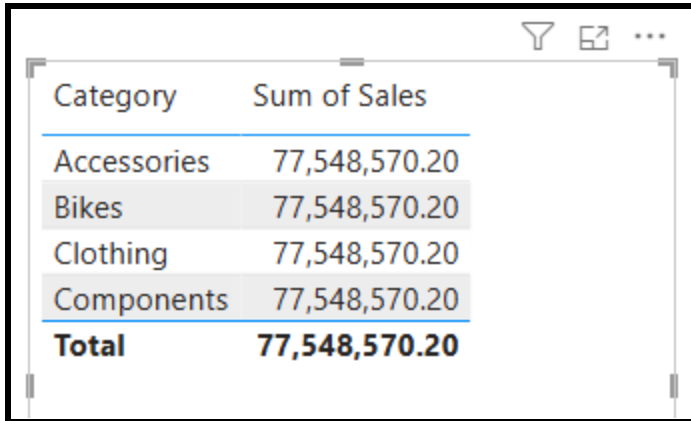
In this task, you'll create model relationships. The file was configured to not identify relationships between tables, which isn't the default setting, but is recommended to prevent extra work creating the correct relationships for your model.

The labs use a shorthand notation to reference a field. It will look like this: Product | Category. In this example, Product is the table name and Category is the field name.

1. In Power BI Desktop, to view all table fields, in the **Data pane**, **right-click an empty area**, and then select **Expand All**.



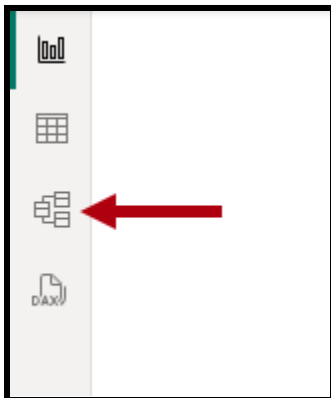
2. From the **Visualization Pane**, select **Table**.
3. To create a table visual, in the Data pane, from inside the **Product table**, check the **Category field**.
4. To add another column to the table, in the Data pane, check the **Sales | Sales field**.
5. Notice that the table visual lists four product categories, and that the sales value is the same for each, and the same for the total.



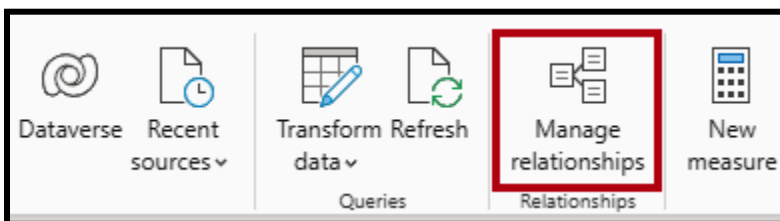
Category	Sum of Sales
Accessories	77,548,570.20
Bikes	77,548,570.20
Clothing	77,548,570.20
Components	77,548,570.20
Total	77,548,570.20

The issue is that the table is based on fields from different tables. The expectation is that each product category displays the sales for that category. However, because there isn't a model relationship between these tables, the Sales table isn't filtered. You'll now add a relationship to propagate filters between the tables.

- To switch to the model designer, at the left select the **Model view icon**.



- On the Home ribbon, select **Manage Relationships**.



- In the Manage Relationships window, notice that no relationships are yet defined.
- To create a relationship, select **+ New relationship**.

10. To configure a relationship from Product table to Sales table, in the From table dropdown list, select the **Product table**, and in the To table dropdown list, select the **Sales table**.

← **New relationship** ×

Select tables and columns that are related.

From table

1 Product

Background C...	Category	Color	Font Color Fo...	Product	ProductKey	Subcatego
#000000	Components	Black	#FFFFFF	HL Road Fra...	210	Road Fra
#000000	Accessories	Black	#FFFFFF	Sport-100 Hel...	215	Helmets
#000000	Accessories	Black	#FFFFFF	Sport-100 Hel...	216	Helmets

To table

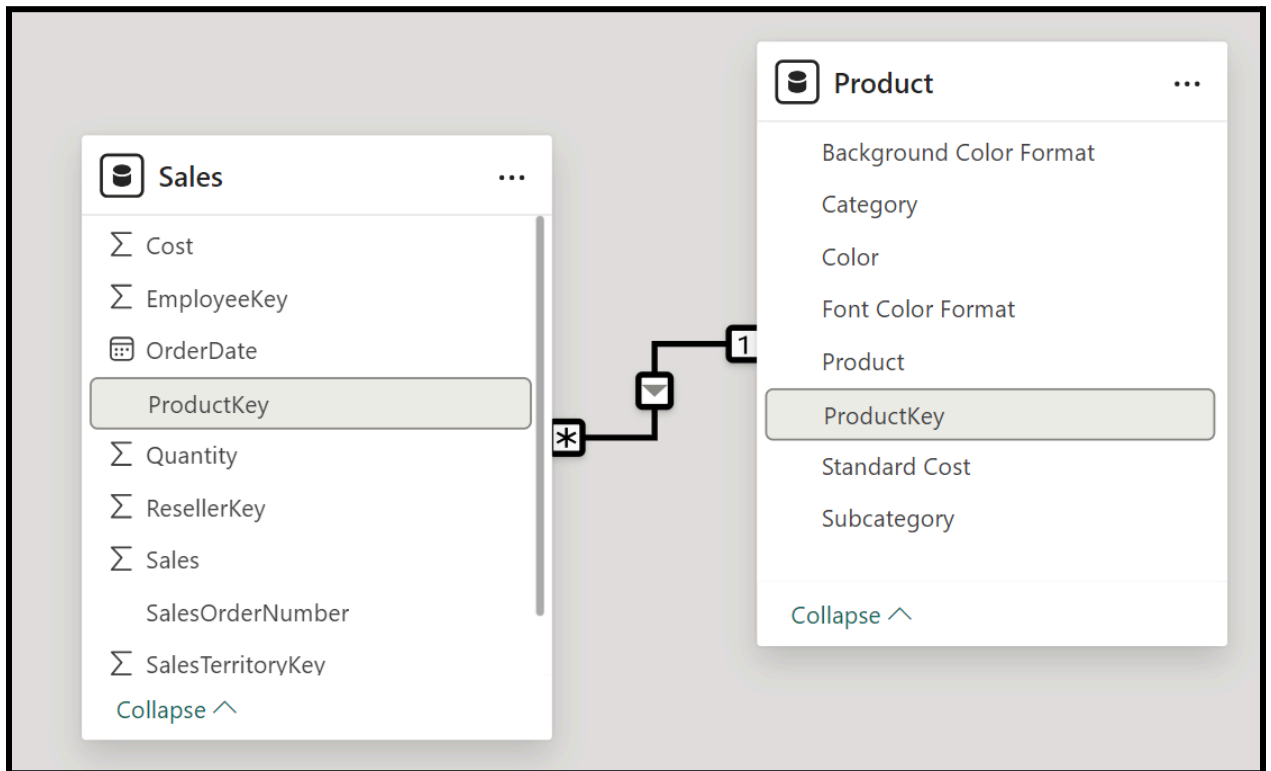
2 Sales

EmployeeKey	OrderDate	ProductKey	Quantity	ResellerKey	SalesOrderNu...	SalesTerrit
282	Friday, 25 Au...	235	2	312	SO43897	4
282	Friday, 25 Au...	351	2	312	SO43897	4
282	Friday, 25 Au...	348	2	312	SO43897	4

11. Notice the following properties were automatically configured:
- **ProductKey columns** in each table are selected.
 - Cardinality type is One To Many (1:*). *The cardinality was automatically detected, because Power BI understands that the **ProductKey column** from the Product table contains **unique values**.*
 - Make **This Relationship Active** is **checked**. *Active relationships propagate filters.*

Cardinality One to many (1:*)	Cross-filter direction Single
☒ Make this relationship active ☐ Assume referential integrity	☐ Apply security filter in both directions

12. Select **Save**, notice in the **Manage Relationships window** that the **new relationship is listed**, and then select **Close**.
13. In the model diagram, notice there's now a connector between the two tables (you might choose to reposition the tables to see the relationship more clearly).



You can interpret many of the relationship properties by looking at the relationship line in the diagram:

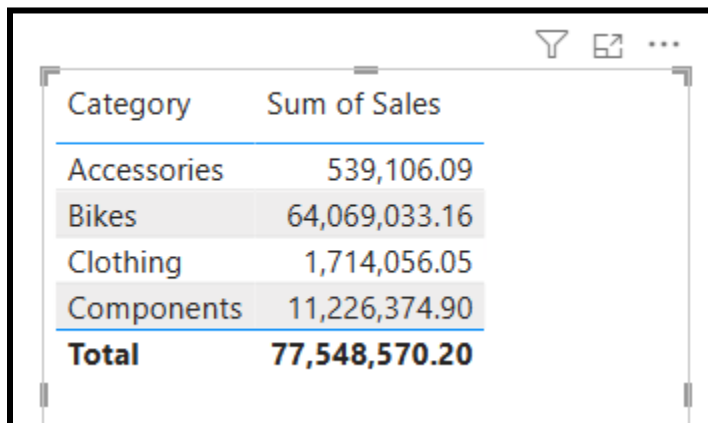
- **Cardinality** is represented by the 1 and (*) indicators.
- **Filter direction** is represented by the arrow head.

- A **solid line** represents an active relationship; a dotted line represents an inactive relationship.

Tip: If you hover the cursor over the relationship to highlight the related columns._

14. Switch to **Report view**, and then notice that the table visual updated to display different values for each product category.

Filters applied to the Product table now propagate to the Sales table.



Category	Sum of Sales
Accessories	539,106.09
Bikes	64,069,033.16
Clothing	1,714,056.05
Components	11,226,374.90
Total	77,548,570.20

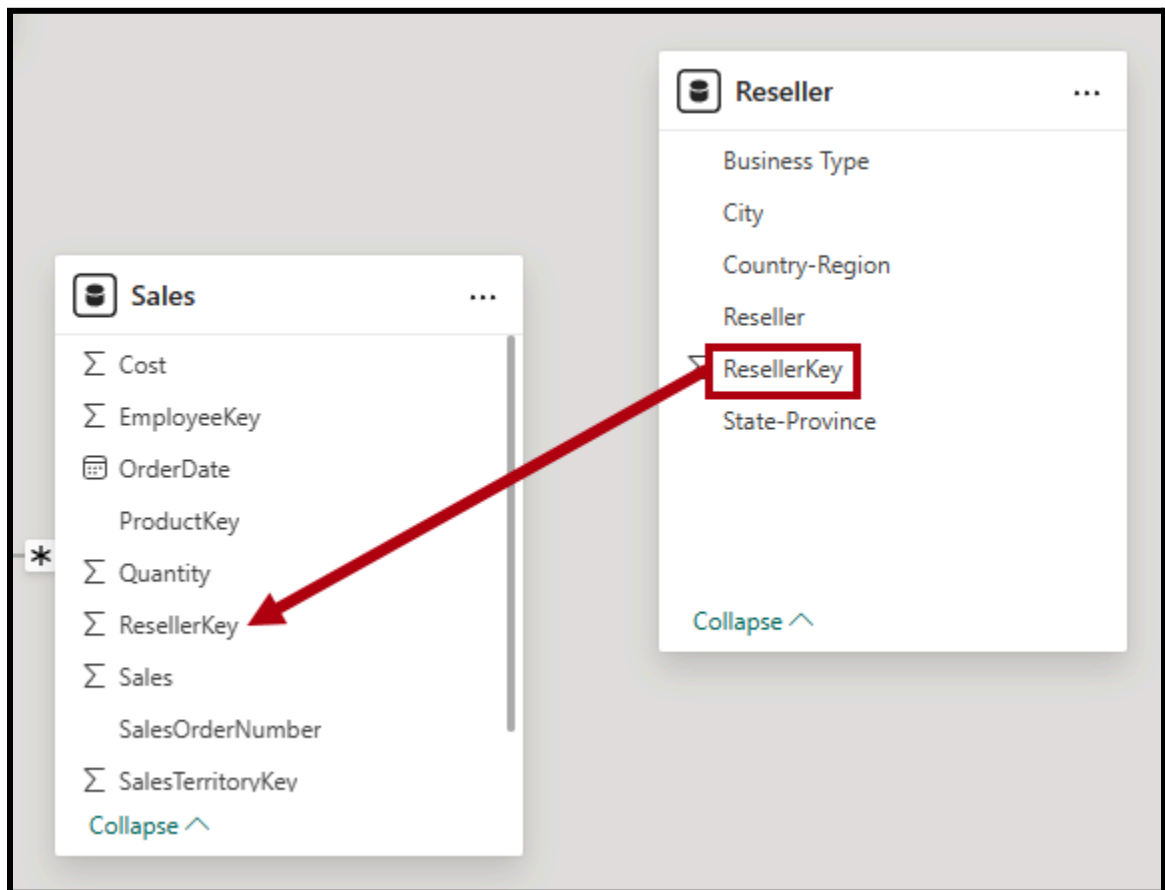
Create additional relationships

There's an easier way to create a relationship. In the model diagram, you can drag and drop columns to create a new relationship.

1. To create a new relationship using a different technique, switch to **Model view**.
2. From the Reseller table, drag the **ResellerKey** column onto the ResellerKey column of the Sales table.

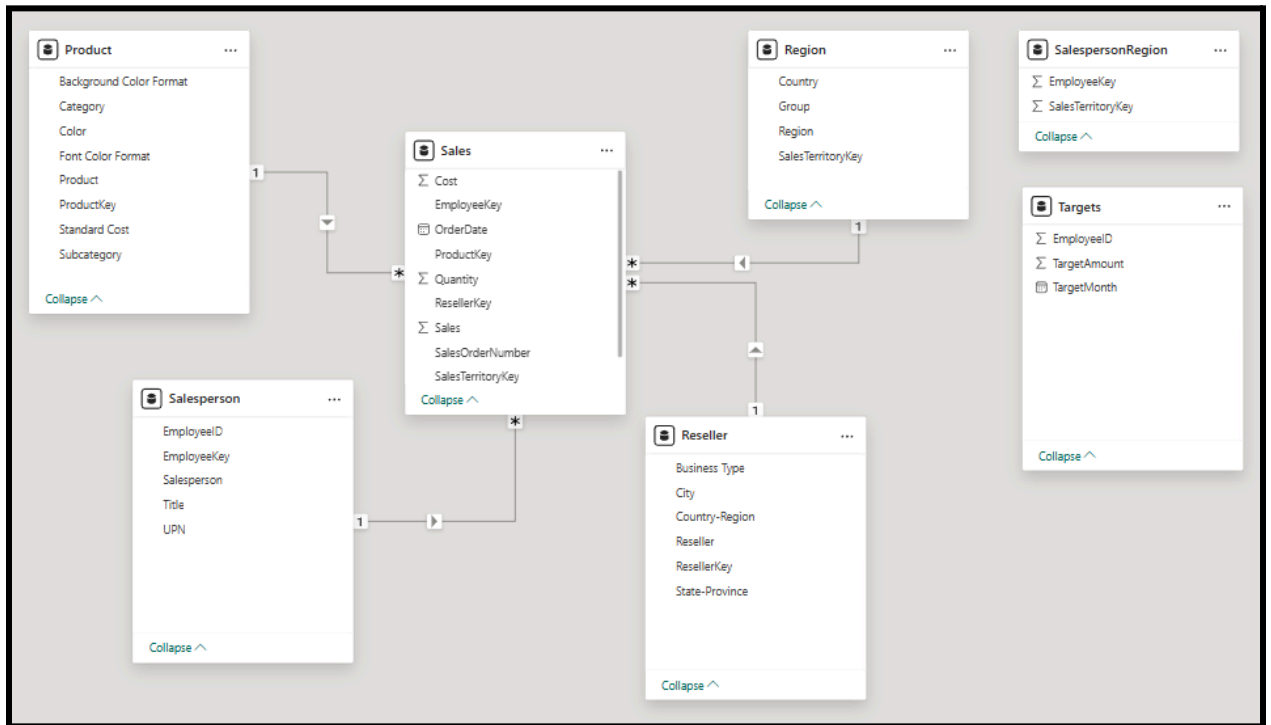
Important: Sometime a column doesn't want to be dragged. If this situation arises, select a different column, and then select the column you intend to drag again, and then try again. Ensure that you

see the new relationship added to the diagram.



3. In the **New relationship window**, review the configuration, and then select **Save**.
4. Use the **new technique** to create the following two model relationships:
 - **Region | SalesTerritoryKey** to Sales | SalesTerritoryKey
 - **Salesperson | EmployeeKey** to Sales | EmployeeKey
5. In the diagram, arrange the tables so that the Sales table is positioned in the center of the diagram, and the related tables are arranged about it. Position

the disconnected tables to the side.



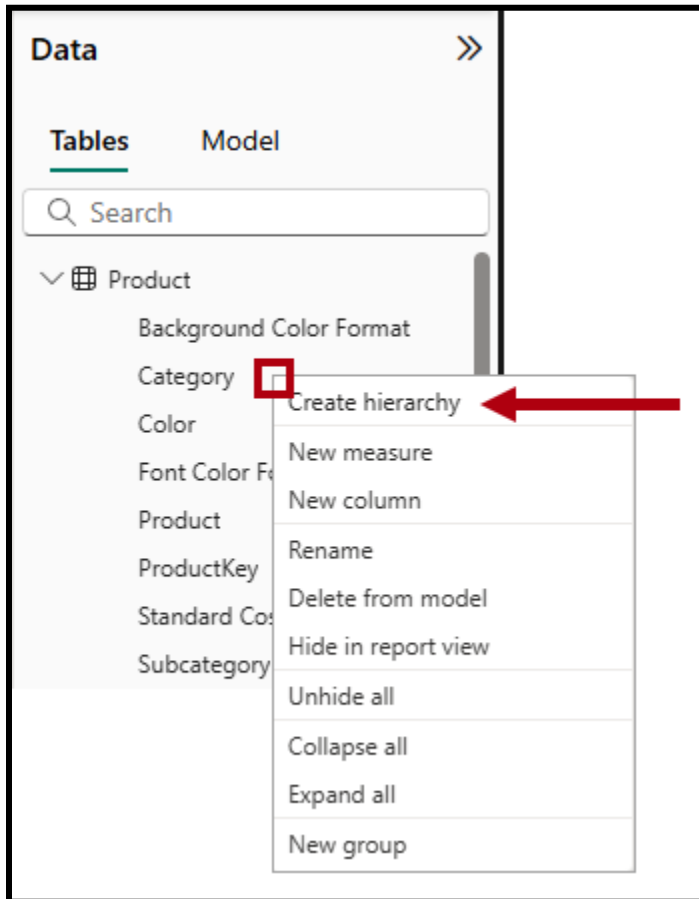
6. **Save** the Power BI Desktop file.

Configure the Product table

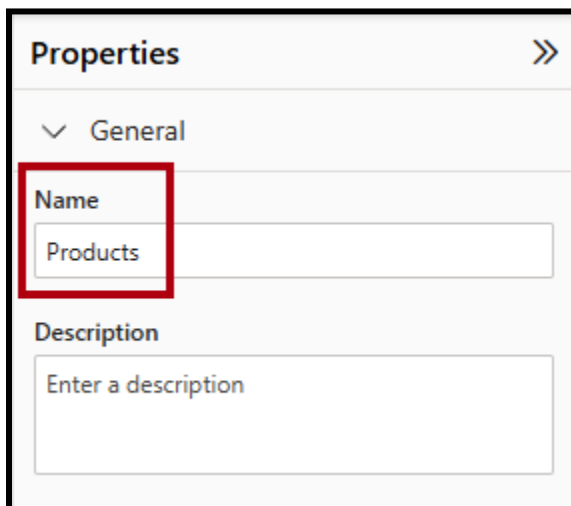
In this task, you'll configure the Product table with a hierarchy and display folder.

1. Switch to **Model view**.
2. In the **Data pane**, if necessary, **expand** the Product table to reveal all fields.

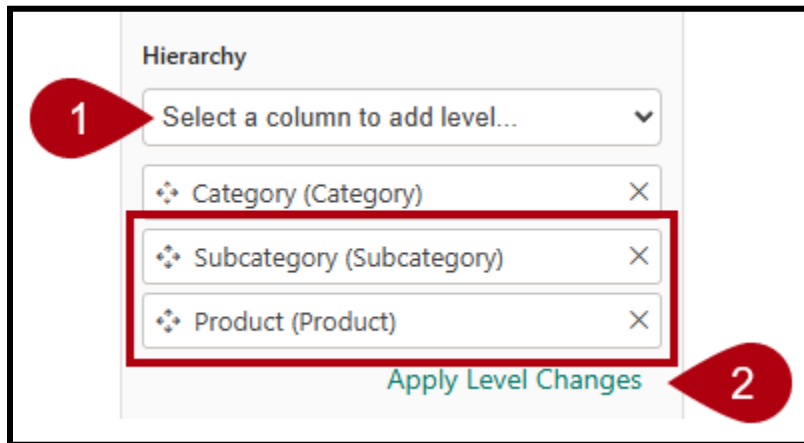
3. In the **Product table**, right-click the **Category** column, and select **Create hierarchy**.



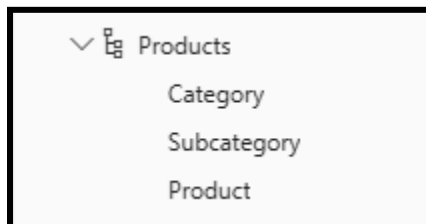
4. In the **Properties** pane, in the Name box, replace the value with *Products*.



5. To add levels to the hierarchy, in the Hierarchy dropdown list, select **Subcategory** and then select **Product**, and then select **Apply Level Changes**.



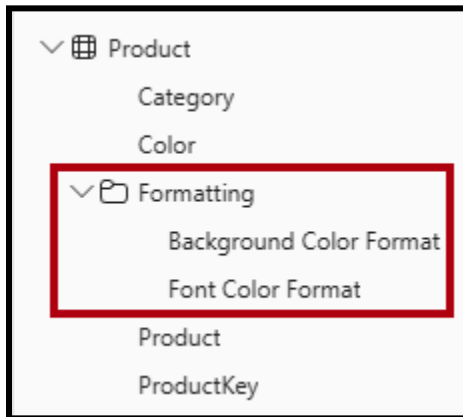
6. In the **Data pane**, notice the Products hierarchy. To reveal the hierarchy levels, expand it.



7. To organize columns into a display folder, in the **Data pane**, select **Product table**, then first select the **Background Color Format column**.
8. While **pressing the Ctrl key**, select the Font Color Format column.
9. In the **Properties pane**, in the Display Folder box, enter **Formatting**.



10. In the **Data pane**, notice that the two columns are now inside a folder.

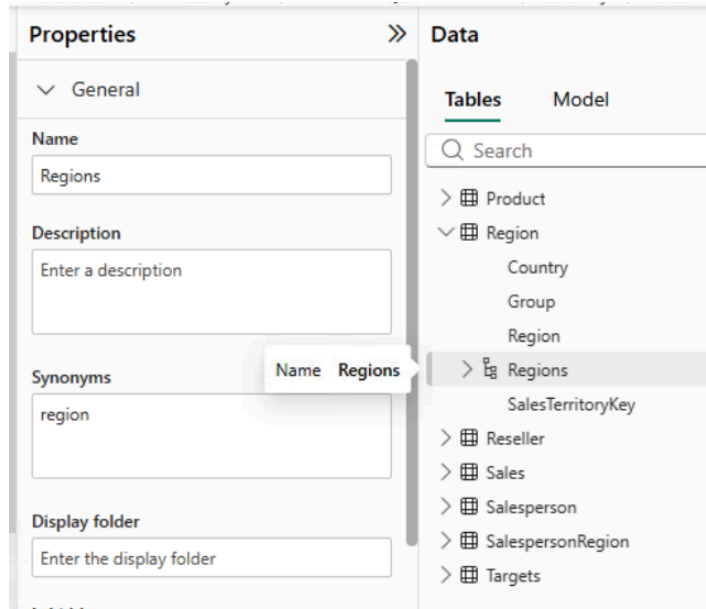


Display folders are a great way to organize tables, especially for tables that comprise many fields. They're logical presentation only.

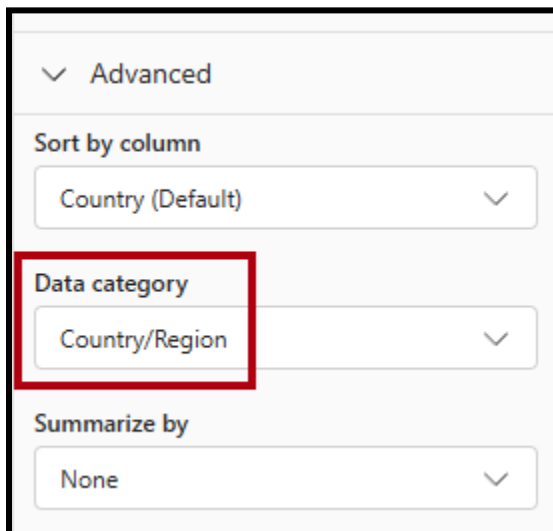
Configure the Region table

In this task, you'll configure the Region table with a hierarchy and updated categories.

1. In the **Region table**, Select **Region** and create a hierarchy named **Regions**, with the following three levels:
 - Group
 - Country
 - Region



2. Select the **Country** column (not the Country hierarchy level).
3. In the **Properties** pane, expand the **Advanced** section (located at the bottom of the pane), and then in the **Data Category** dropdown list, select **Country/Region**.

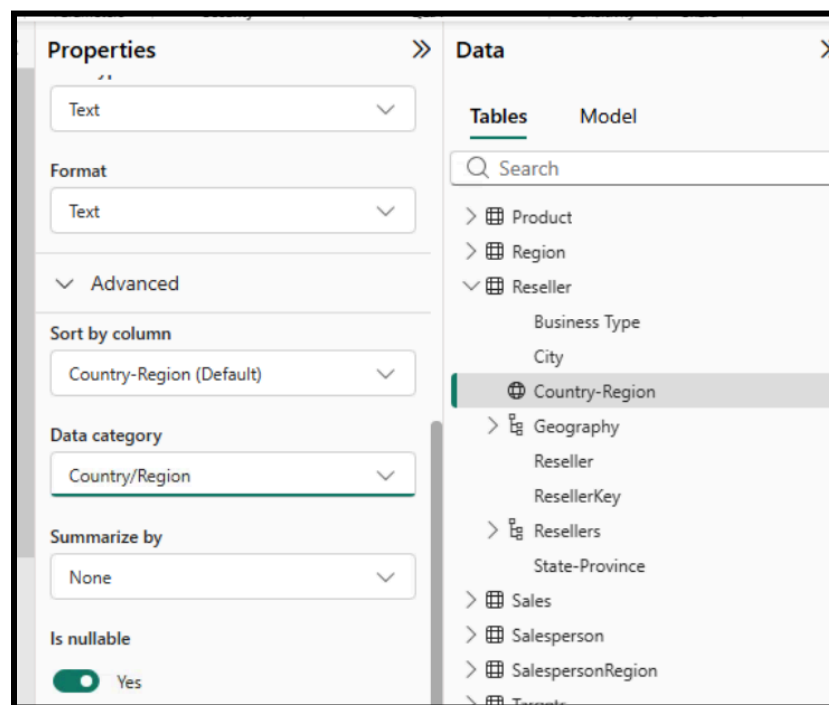


*Data categorization can provide hints to the report designer. In this case, categorizing the column as country or region provides more accurate information to Power BI when it renders a **map** visualization.*

Configure the Reseller table

In this task, you'll configure the Reseller table to add a hierarchy and update data categories.

1. In the **Reseller table**, create a hierarchy named **Resellers**, with the following two levels:
 - Business Type
 - Reseller
2. Create a **second hierarchy** named **Geography**, with the following four levels:
 - Country-Region
 - City
 - Reseller
 - Reseller
3. Set the **data category** for the following columns (not within the hierarchy):
 - Country-Region to Country/Region



Configure the Sales table

In this task, you'll configure the **Sales table** with updated descriptions, formatting, and summarization.

1. In the **Sales table**, select the **Cost** column.
2. In the **Properties pane**, in the Description box, **enter**: *Based on standard cost*
Descriptions can be applied to tables, columns, hierarchies, or measures. In the Data pane, description text is revealed in a tooltip when a report author hovers their cursor over the field.
3. Select the Quantity column.
4. In the **Properties pane**, from inside the Formatting section, set the **Thousands Separator** property to **Yes**.
5. Select the Unit Price column.
6. In the Properties pane, from inside the Formatting section, set the **Decimal Places** property to **2**.
7. In the Advanced group (you may need to scroll down to locate it), in the Summarize By dropdown list, select **Average**.

By default, numeric columns will summarize by summing values together. This default behavior isn't suitable for a column like Unit Price, which represents a rate. Setting the default summarization to average will produce a meaningful result.

Bulk update properties

In this task, you'll update **multiple columns using single bulk updates**. You'll use this approach to hide columns, and format column values.

1. In the **Data pane** (or model diagram), select the Product | ProductKey column.
2. While pressing the **Ctrl key**, select the following 13 columns (spanning multiple tables):
 - Region | SalesTerritoryKey

- Reseller | ResellerKey
- Sales | EmployeeKey
- Sales | ProductKey
- Sales | ResellerKey
- Sales | SalesOrderNumber
- Sales | SalesTerritoryKey
- Salesperson | EmployeeID
- Salesperson | EmployeeKey
- Salesperson | UPN
- SalespersonRegion | EmployeeKey
- SalespersonRegion | SalesTerritoryKey
- Targets | EmployeeID

3. In the Properties pane, set the **Is Hidden property** to **Yes**.

The columns were hidden because they're either used by relationships or will be used in row-level security configuration or calculation logic.

*You'll use the **SalesOrderNumber** column in a calculation in the Create DAX Calculations in Power BI Desktop lab.*

4. Multi-select the following three columns:

- Product | Standard Cost
- Sales | Cost
- Sales | Sales

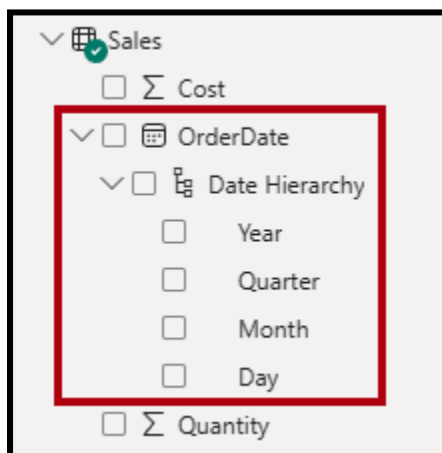
5. In the **Properties pane**, from inside the Formatting section, set the **Decimal Places property** to **0** (zero).

Explore the model interface

In this task you'll switch to **Report view**, review the data model interface, and configure the **auto date/time setting**.

1. Switch to **Report view**.
2. In the **Data pane**, notice the following:

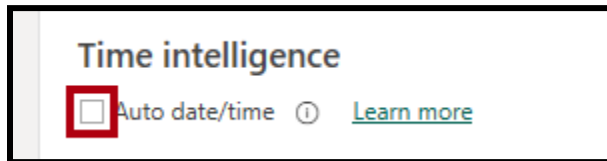
- Columns, hierarchies and their levels are fields, which can be used to configure report visuals.
 - Only fields relevant to report authoring are visible.
 - The **SalespersonRegion** table isn't visible—because all of its fields are hidden.
 - Spatial fields in the **Region and Reseller table** are adorned with a spatial icon.
 - Fields adorned with the sigma symbol (Σ) will summarize, by default.
 - A tooltip appears when hovering the cursor over the **Sales | Cost field**.
3. **Expand** the Sales | OrderDate field, and then notice that it reveals a Date Hierarchy. The Targets | TargetMonth field delivers a similar hierarchy.



Important: These hierarchies weren't created by you. They were created automatically as a default setting. There's a problem, however. The Adventure Works financial year commences on July 1 of each year. But, in these automatically created date hierarchies, the date hierarchy year commences on January 1 of each year.

4. **To turn off the auto date/time setting**, navigate to File > Options and Settings > Options.

5. In the **Options window**, on the Current File section, navigate to **Data Load > Time Intelligence**, and **uncheck** Auto Date/Time.

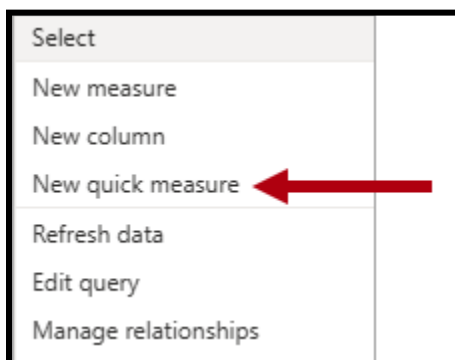


6. In the **Data pane**, notice that the date hierarchies are no longer available.

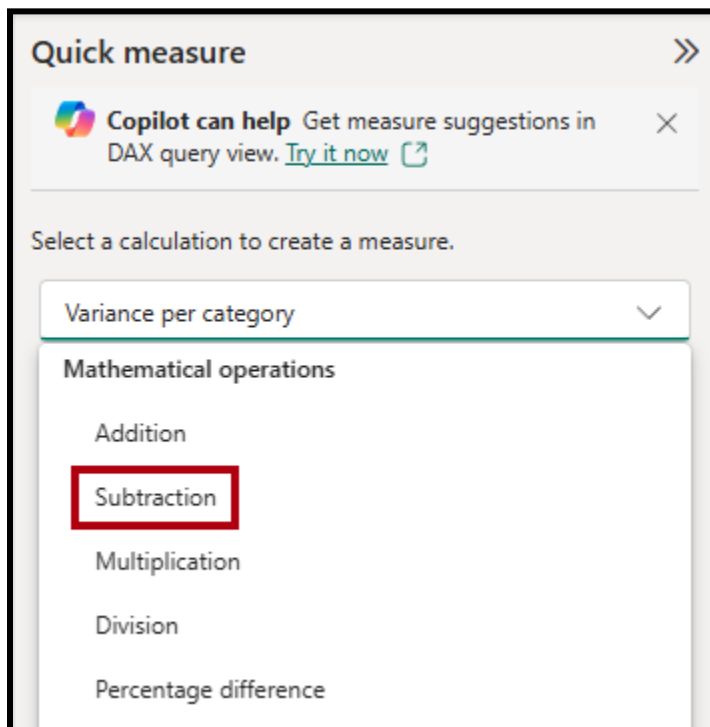
Create quick measures

In this task, you'll create two quick measures to calculate **profit and profit margin**. A quick measure creates the calculation formula for you. They're easy and fast to create for simple and common calculations.

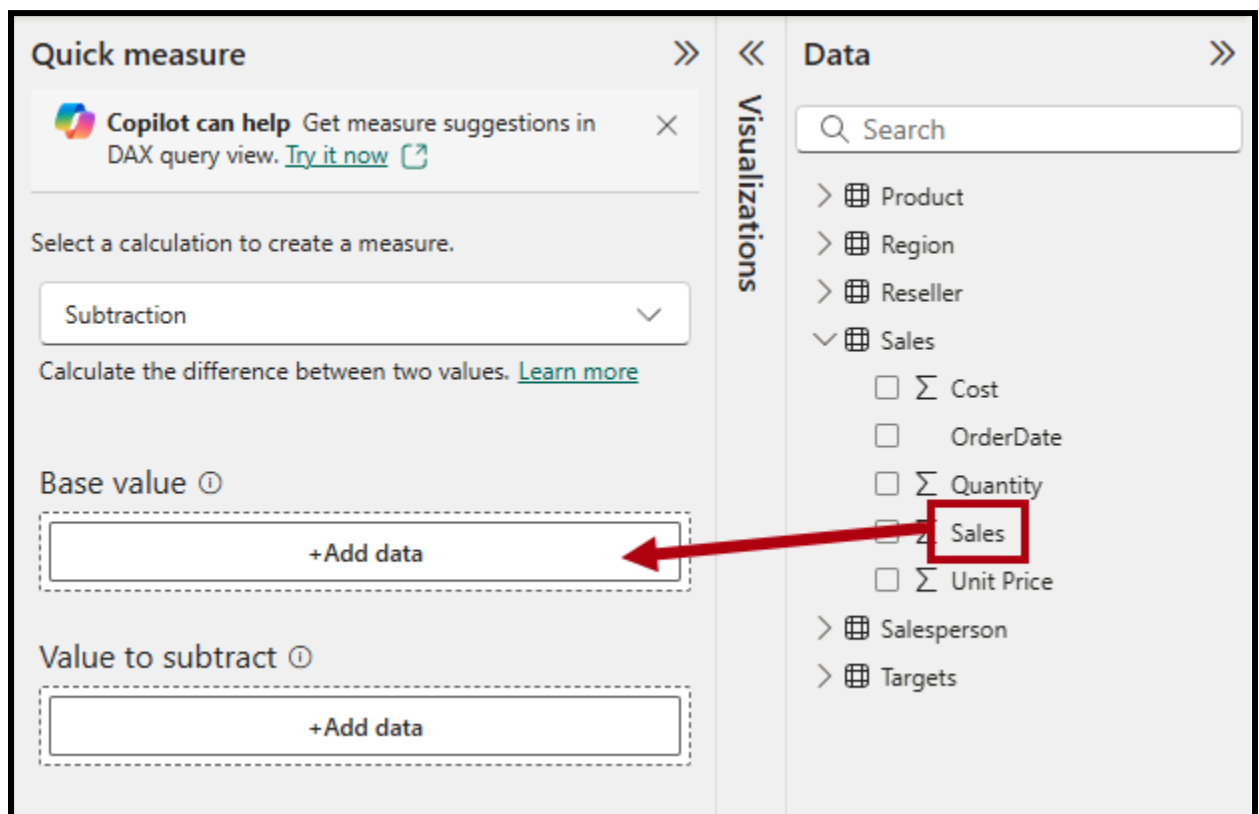
1. In the **Data pane**, right-click the **Sales table**, and then select **New Quick Measure**.



2. In the Quick Measure pane, in the **Select a Calculation dropdown list**, from inside the Mathematical Operations group, select **Subtraction**.



3. From the **Data pane**, drag the Sales | Sales field into the Base Value well.



4. Drag the Sales | Cost field into the Value to Subtract box.

Base value ⓘ

1 Sum of Sales X | >

Value to subtract ⓘ

2 Sum of Cost X | >

5. Select **Add**.

6. In the **Data pane**, inside the **Sales table**, notice that new measure.
Measures are indicated by the calculator icon.

▼ Sales

- ☐ Σ Cost
- ☐ OrderDate
- ☐ Σ Quantity
- ☐ Σ Sales
- ☒ Sum of Sales minus Sum of Cost
- ☐ Σ Unit Price

7. To **rename** the measure, right-click it, select Rename, then rename as **Profit**.
Tip: To rename a field, you can also double-click it, or select it and press F2.

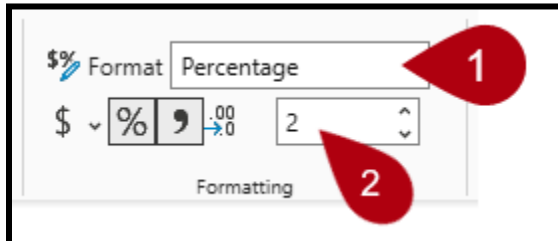
8. In the **Sales table**, add a second quick measure, based on the following requirements:

_Important: If the option to create a quick measure doesn't appear in the context menu, use the command located on the Home ribbon, from inside the Calculations group.

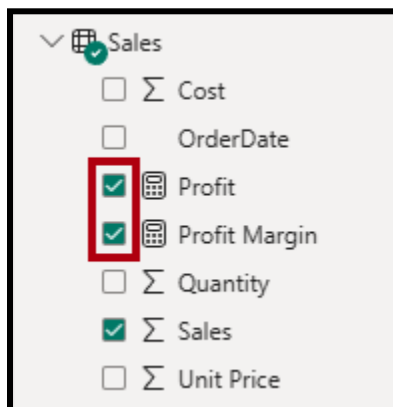
- Use the **Division mathematical operation**.
- Set the **Numerator** to the Sales | Profit field.
- Set the **Denominator** to Sales | Sales field.

- **Rename** the measure as *Profit Margin*.

9. Ensure the **Profit Margin measure** is selected, and then on the Measure Tools contextual ribbon, set the format to **Percentage**, with **two decimal places**.



10. To test the two measures, first select the **existing table** visual on the page.
11. In the **Data pane**, check the Profit and Profit Margin measures to add them to the table.



12. **Select and drag** the right guide to widen the table visual.

Category	Sales	Profit	Prc
Accessories	\$539,106	\$188,081	
Bikes	\$64,069,033	(\$435,679)	
Clothing	\$1,714,056	\$245,857	
Components	\$11,226,375	\$1,001,235	
Total	\$77,548,570	\$999,495	

13. Verify that the measures produce reasonable results that are correctly formatted.

Category	Sales	Profit	Profit Margin
Accessories	\$539,106	\$188,081	34.89 %
Bikes	\$64,069,033	(\$435,679)	-0.68 %
Clothing	\$1,714,056	\$245,857	14.34 %
Components	\$11,226,375	\$1,001,235	8.92 %
Total	\$77,548,570	\$999,495	1.29 %

Create a many-to-many relationship

In this task, you'll create a many-to-many relationship between the Salesperson table and the Sales table.

1. In **Report view**, select a blank area of the report page.
2. To create a new table visual, in the **Data pane**, check the following two fields:
 - Salesperson | Salesperson
 - Sales | Sales

Salesperson	Sum of Sales
Amy Alberts	\$737,568
David Campbell	\$3,614,761
Garrett Vargas	\$3,486,102
Jae Pak	\$8,099,817
Jillian Carson	\$9,755,992
José Saraiva	\$5,536,439
Linda Mitchell	\$10,158,635
Lynn Tsoflias	\$1,216,917
Michael Blythe	\$8,952,751
Pamela Ansman-Wolfe	\$3,187,720
Rachel Valdez	\$1,681,208
Ranjit Varkey Chudukatil	\$4,429,368
Shu Ito	\$6,283,193
Stephen Jiang	\$1,073,651
Syed Abbas	\$174,108
Tete Mensa-Annan	\$2,184,211
Tsvi Reiter	\$6,976,128
Total	\$77,548,570

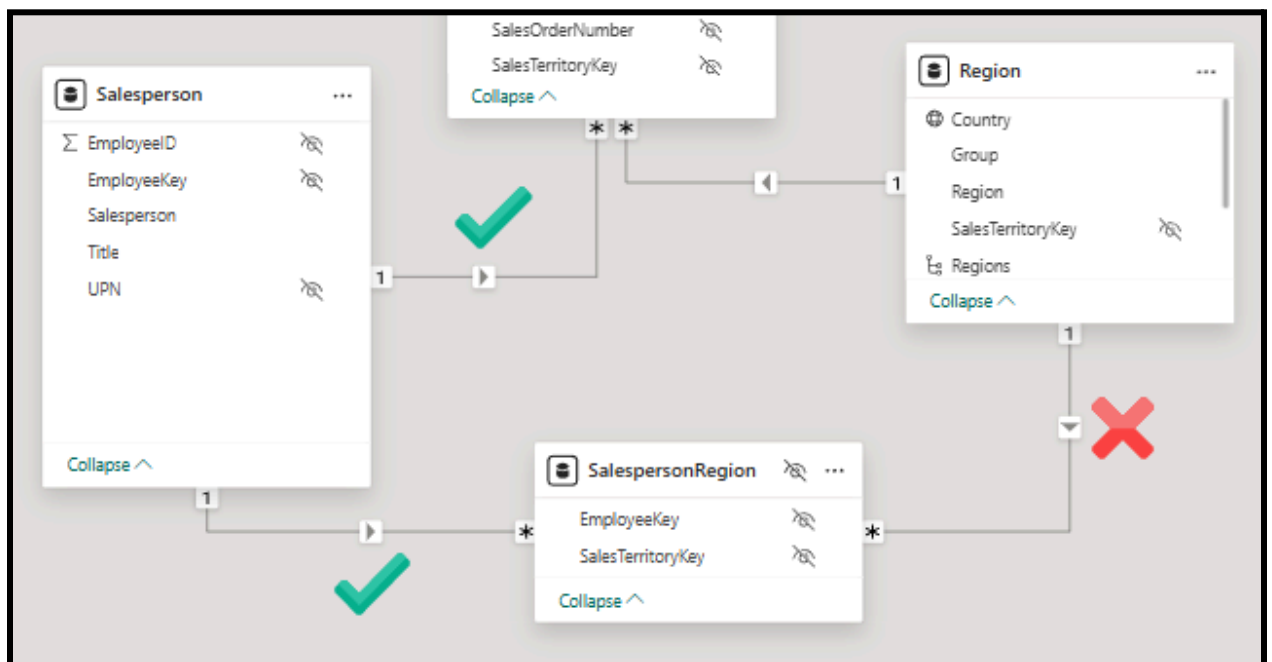
3. The table visual displays sales made by each salesperson. However, there's another relationship between salespeople and sales. Some salespeople belong to one, two, or possibly more sales regions. In addition, sales regions can have multiple salespeople assigned to them.

From a performance management perspective, a salesperson's sales (based on their assigned territories) need to be analyzed and compared with sales targets. You'll create relationships to support this analysis in the next exercise.

4. Notice that *Michael Blythe* has generated almost 9 million dollars of sales.
5. Switch to **Model view**, then drag the **SalespersonRegion table** to position it between the Region and Salesperson tables.
6. Use the **drag-and-drop** technique to create the following two model relationships:
 - Salesperson | EmployeeKey to SalespersonRegion | EmployeeKey

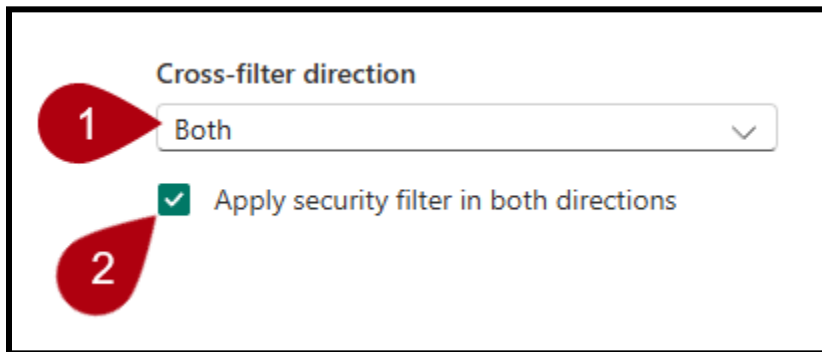
- Region | SalesTerritoryKey to SalespersonRegion | SalesTerritoryKey

7. The **SalespersonRegion** table can be considered to be a bridging table.
8. Switch to **Report view**, and then notice that the visual hasn't updated—the sales result for Michael Blythe hasn't changed.
9. Switch back to **Model view**, and then follow the relationship filter directions (arrowhead) from the Salesperson table.
Consider that the Salesperson table filters the Sales table. It also filters the SalespersonRegion table, but it doesn't continue by propagating filters to the Region table (the arrowhead is pointing the wrong direction).



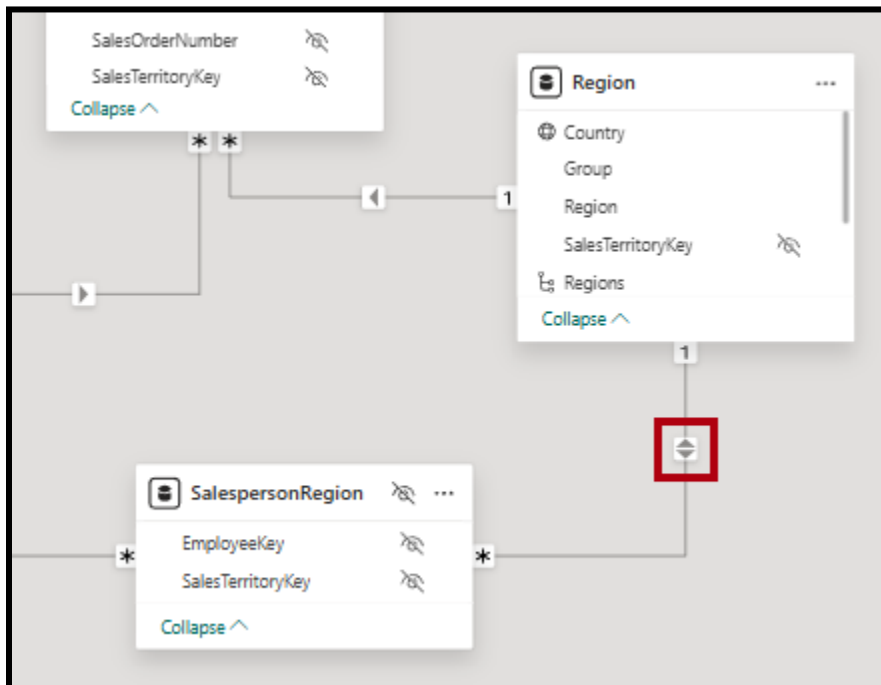
10. To edit the relationship between the **Region** and **SalespersonRegion** tables, double-click the relationship.
11. In the Edit Relationship window, in the Cross Filter Direction dropdown list, select **Both**.

12. Check the Apply Security Filter in Both Directions checkbox.



13. Select **Save**.

14. Notice that the relationship has a double arrowhead now.

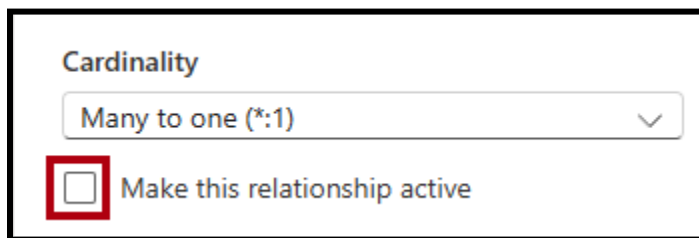


15. Switch to **Report view**, and then notice that the sales values have still not changed.

The issue now relates to the fact that there are two possible filter propagation paths between the Salesperson and Sales tables. This ambiguity is internally resolved, based on a "least number of tables" assessment. To be clear, you shouldn't design models with this type of ambiguity—the issue will be

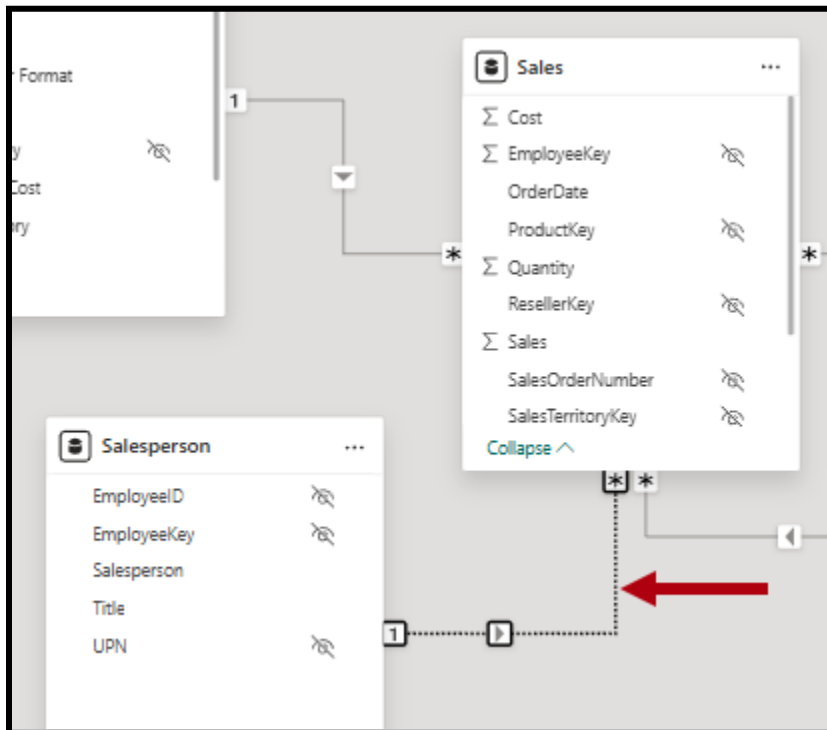
addressed in part later in this lab, and by the completion of the Create DAX Calculations in Power BI Desktop lab.

16. Switch to **Model view**.
17. To force filter propagation via the bridging table, edit (double-click) the relationship between the **Salesperson** and **Sales tables**.
18. In the Edit Relationship window, uncheck the Make This Relationship Active checkbox.



19. Select **Save**.
Filter propagation will now follow the only active path.

20. In the model diagram, notice that the **inactive relationship** is represented by a dotted line.



21. Switch to **Report view**, and then notice that the sales for Michael Blythe are now nearly 22 million dollars.

22. Notice also, that the sales for each salesperson—if added—would exceed the table total.

It's a common observation of a many-to-many relationship due to the double, triple, etc. counting of regional sales results. Consider Brian Welcker, the second salesperson listed. His sales amount equals the total sales amount. It's the correct result due to the fact that he's the Director of Sales; his sales are measured by the sales of all regions.

While the many-to-many relationship is now working, it's now not possible to analyze sales made by a salesperson (because the relationship is inactive). You'll be able to reactivate the relationship when you introduce a calculated table that will allow analyzing sales made in the sales region(s) assigned to the salesperson (for performance analysis) in the Create DAX Calculations in Power BI Desktop lab.

23. Switch to **Model view**, and then in the model diagram, select the **Salesperson table**.

24. In the **Properties** pane, in the Name box, replace the text with **Salesperson (Performance)**.

The renamed table now reflects its purpose: it's used to report and analyze the performance of salespeople based on the sales of their assigned sales regions.

Relate the Targets table

In this task, you'll create a relationship to the Targets table.

1. Create a relationship from the **Salesperson (Performance) | EmployeeID** column and the **Targets | EmployeeID** column.
2. In **Report view**, add the **Targets | Target** field to the table visual.
3. Resize the table visual so all columns are visible.

Salesperson	Sum of Sales	Sum of Target
Amy Alberts	\$10,288,626	\$19,450,000
Brian Welcker	\$77,548,570	\$221,700,000
David Campbell	\$12,004,822	\$19,625,000
Garrett Vargas	\$13,875,633	\$23,675,000
Jae Pak	\$8,410,883	\$13,575,000
Jillian Carson	\$7,633,387	\$13,675,000
José Saraiva	\$13,875,633	\$18,875,000
Linda Mitchell	\$25,634,503	\$40,850,000
Lynn Tsoflias	\$1,391,025	\$3,210,000
Michael Blythe	\$21,987,348	\$31,150,000
Pamela Ansman-Wolfe	\$30,005,939	\$53,850,000
Rachel Valdez	\$1,877,743	\$4,125,000
Ranjit Varkey Chudukatil	\$4,527,840	\$9,050,000
Shu Ito	\$18,001,116	\$59,850,000
Stephen Jiang	\$65,868,919	\$110,150,000
Syed Abbas	\$1,391,025	\$3,050,000
Tete Mensa-Annan	\$12,004,822	\$17,100,000
Tsvi Reiter	\$7,638,607	\$13,250,000
Total	\$77,548,570	\$676,210,000

It's now possible to visualize sales and targets—but for now take care for two reasons. First, there's no filter on a time period, and so targets also include future target amounts. Second, targets aren't additive, and so the total shouldn't be displayed. They can either be disabled by formatting the visual or removed by using calculation logic.

Save the Power BI Desktop file.

Lab complete

You may choose to save your Power BI report, though it's not necessary for this lab. In the next exercise, you'll work with a pre-made starter file.

1. Navigate to the "File" menu in the top left corner and select "Save As".
2. Select **Browse this device**.
3. Select the folder where you want to save the file and give it a descriptive name.
4. Select the Save button to save your report as a .pbix file.
5. If a dialog box appears prompting you to apply pending query changes, select Apply.
6. Close Power BI Desktop.